UK NUCLEAR POWER SAFETY VERIFICATION FRAMEWORK

MSc Research Technical Report & System Architecture Document (UK ONR & IAEA Scope)

1. Executive Summary

This research framework presents a multi-modal decision support platform for verifying statements concerning UK nuclear power safety (covering Hinkley Point C, Sizewell B/C, Torness, Heysham, Dungeness B, and Sellafield). Operating under UK Office for Nuclear Regulation (ONR) guidelines and IAEA international standards, the system combines deep transformer sequence classification with a transparent **0–100 Multi-Factor Trust Engine** across **3 input modalities** (raw text, direct article URLs, and news graphics/screenshots via deep-learning OCR).

2. System Methodology & Core Components

- **Natural Language Processing (NLP):** Fine-tuned transformer models evaluate subtle contextual syntactic markers distinguishing authentic technical reporting from deceptive claims.
- **Supervised Transfer Learning:** Sequence classifiers are initialized from domain-specific checkpoints and tuned using class-weighted cross-entropy loss and label smoothing.
- **Computer Vision OCR:** Deep-learning Optical Character Recognition (EasyOCR) extracts text from social media graphics and article screenshots.

3. Models & Decision Components in the System

Component Name	Type & Framework	Role & Technical Explanation	Data Consumed
Domain Relevance Filter	Regex Keyword Matching Engine	Nuclear Topic Checker: Verifies if incoming text concerns nuclear power/energy. Out-of-domain news is labeled 'Not Related to Nuclear Power'.	Input article text & headline tokens.
RoBERTa / DeBERTa	Transformer Deep Learning (PyTorch)	Main Text Classifier: Evaluates linguistic context to detect subtle fake news patterns versus authentic technical reporting.	Fine-tuned on UK & IAEA nuclear safety dataset.
EasyOCR Engine	Computer Vision OCR (PyTorch + OpenCV)	Image Text Extractor: Reads printed text from uploaded news screenshots and social media graphics.	Raw image pixels (PNG, JPG, WebP screenshots).
VADER Sentiment Analyzer	NLP Sentiment & Tone Evaluator	Emotional Tone Checker: Measures whether article phrasing is calm and objective or overly sensational/alarmist.	Parsed article text & headline tokens.
Publisher Credibility Evaluator	Heuristic Rule Engine	Publisher Verifier: Assigns credibility rankings based on domain authority (e.g., GOV.UK, BBC News, ONR vs clickbait blogs).	URL domain names & publisher headers.
Official Corroboration Engine	Keyword Overlap Matcher	Official Log Cross-Checker: Verifies statements against authenticated UK ONR and IAEA nuclear safety incident databases.	IAEA IEC, UK ONR & EURDEP safety claims.
Multi-Factor Trust Engine	Ensemble Decision Engine	Score Calculator: Combines model confidence, publisher trust, corroboration, and tone into a final 0–100 Trust Score.	Outputs from all sub-models above.

4. Datasets & Domain Scope

- **Primary Fine-Tuning Dataset (data/sample_dataset.csv):** 654 unique rows focusing on UK nuclear power stations (*Hinkley Point C, Sizewell B/C, Torness, Heysham 1/2, Dungeness B, Sellafield, Hunterston B, Wylfa*) and IAEA standards.

- **Held-Out Test Dataset (data/holdout_test.csv):** 30 hand-crafted test statements never seen during training to evaluate real-world generalization (achieving 93% accuracy).
- **Data Specificity:** Pure UK and IAEA scope with zero non-UK or Indian data included.
- **Out-of-Domain Handling:** Non-nuclear articles (e.g. general sports, politics) are safely identified as 'Not Related to Nuclear Power'.

5. Data Sources Specific to Nuclear Fact-Checking

1. **IAEA Incident and Emergency Centre (IEC) Reports:** Official international bulletins for radiological incidents.
2. **National Regulators:** UK Office for Nuclear Regulation (ONR), UKAEA, DESNZ / GOV.UK, and US NRC declarations.
3. **Radiation Monitoring Networks:** EURDEP (European Radiological Data Exchange Platform) & EPA RadNet.
4. **Nuclear Fact-Checking Archives:** Verified tagged archives from Full Fact UK, Snopes, and PolitiFact.

6. Handle Uncertainty Honestly (Section 7 Policy)

For high-consequence topics like nuclear safety, when a statement has no corroborating match in official logs and lacks clickbait/alarmist markers, the system **does not force a fake score**. It flags the item as **'Insufficient Verification Data'** (Trust Score 50–65/100) to prevent dangerous false positives on genuine safety information.

7. Trust Score Calculation Weights

- **35% Model Confidence:** Transformer classifier prediction probability.
- **25% Source Credibility:** Reputation ranking of the news domain/source.
- **25% Fact Corroboration:** Match against verified IAEA IEC and UK ONR safety logs.
- **15% Tone Neutrality:** Absence of clickbait / sensational emotional language.